Linear mixed effects models

Last time

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

where $Price_{ij}$ is the price of rental j in neighborhood i

- u_i is a *random intercept*
- We use subscripts i and j for $Price_{ij}$, $Satisfaction_{ij}$, and ε_{ij} because they are different for every observation in the data
- We only need a subscript i (neighborhood) for u_i, because there is one random intercept per neighborhood

Last time: Class activity

```
Fixed effects:
                       Estimate Std. Error t value
(Intercept)              27.275     30.547   0.893
overall_satisfaction     14.809      6.303   2.350
```

Interpret the estimate fixed effect coefficients $\widehat{\beta}_0$ and $\widehat{\beta}_1$.

Last time: Class activity

```
Random effects:
 Groups       Name        Variance Std.Dev.
 neighborhood (Intercept) 1048     32.37
 Residual                 6762     82.23
```

> Calculate and interpret the estimated intra-class correlation.

Warmup

Work on the warmup activity (handout).

Warmup

> What are the groups in the data?

Warmup

> What mixed effects model should the researchers use to investigate their hypothesis?

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$u_i \overset{iid}{\sim} N(0, \sigma_u^2)$ $\quad$ $\varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$

where $Price_{ij}$ is the price of rental j in neighborhood i

> What assumptions are we making in this mixed effects model?

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Shape:**
  - the overall relationship between satisfaction and price is linear
  - The slope is the *same* for each neighborhood

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Independence:**
  - random effects are independent
  - observations within neighborhoods are independent after accounting for the random effect (i.e., the random effect captures the correlation within neighborhoods)
  - observations from different neighborhoods are independent

Assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Normality:** Both $u_i \sim N(0, \sigma_u^2)$ and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$
- **Constant variance:**
  - ε_{ij} has the same variance σ_ε^2 regardless of satisfaction or neighborhood

Checking assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

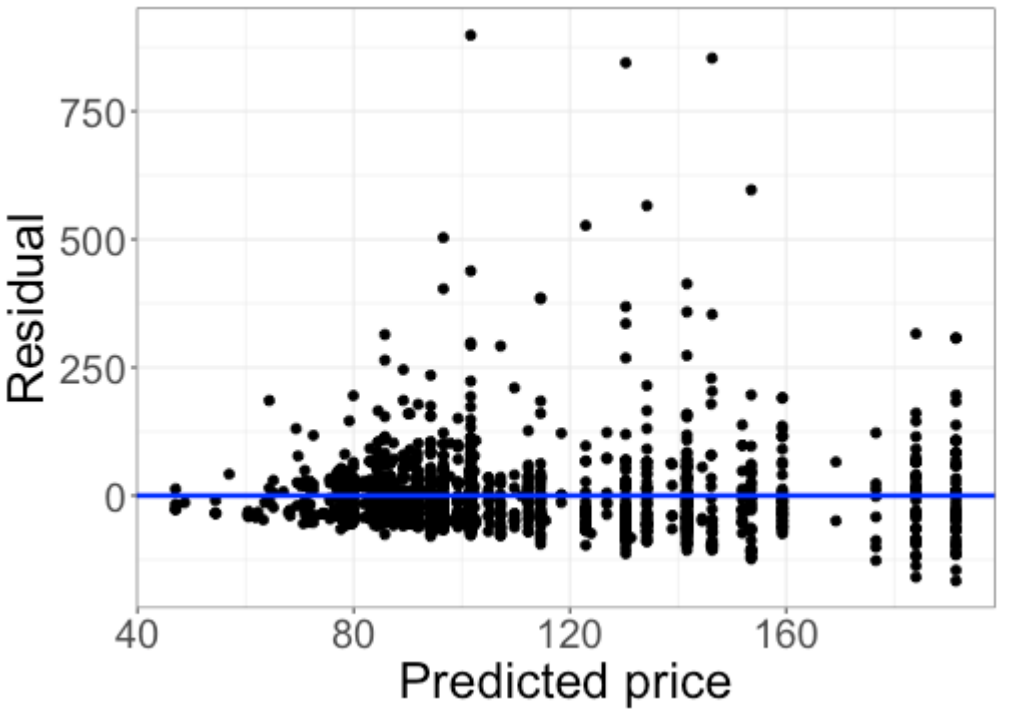
- **Shape assumption:**
  - the overall relationship between satisfaction and price is linear
  - The slope is the *same* for each neighborhood
- **Constant variance assumption:**
  - ε_{ij} has the same variance σ_ε^2 regardless of satisfaction or neighborhood

> How do you think we could check the shape and constant variance assumptions?

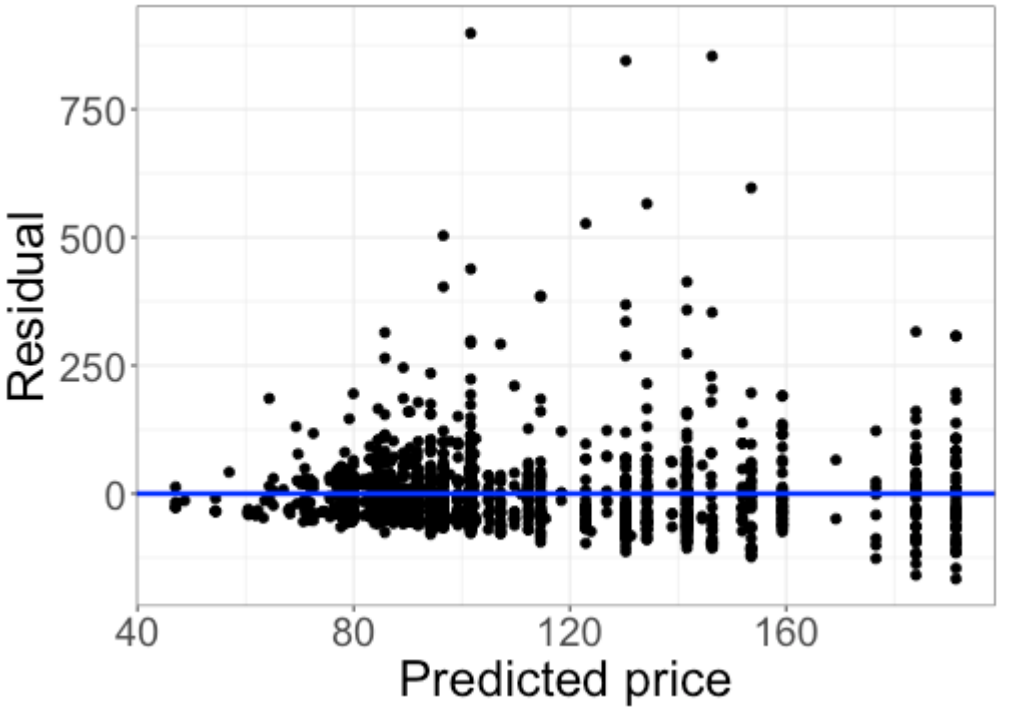
Residual plots

Residuals: $Price_{ij} - \widehat{Price}_{ij}$, where

$$\widehat{Price}_{ij} = \widehat{\beta}_0 + \widehat{\beta}_1 Satisfaction_{ij} + \widehat{u}_i$$

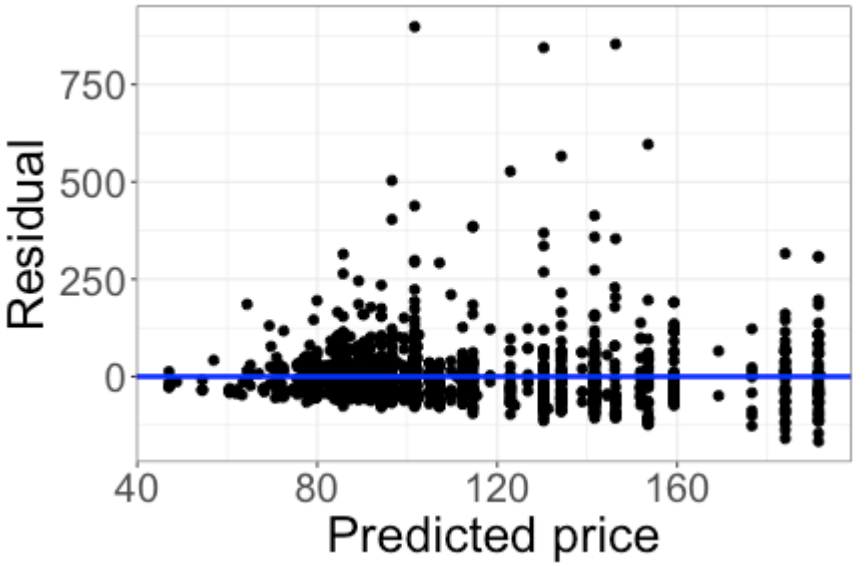


Residual plots



Do the shape and constant variance assumptions look reasonable?

Residual plots



- Linearity seems reasonable
- We can't assess (yet) whether the slope is actually the same for all neighborhoods
- Constant variance may be slightly violated (but probably not enough to matter)

Calculating predicted values

Residuals: $Price_{ij} - \widehat{Price}_{ij}$, where

$$\widehat{Price}_{ij} = \widehat{\beta}_0 + \widehat{\beta}_1 Satisfaction_{ij} + \widehat{u}_i$$

> Where do the $\widehat{u}_i$ come from?

Estimated random effects

R calculates an estimated random effect for each group (i.e., neighborhood):

```
m1 <- lmer(price ~ overall_satisfaction +
             (1 | neighborhood),
           data = bnb)
coef(m1)
```

```
## $neighborhood
##                   (Intercept) overall_satisfaction
## Albany Park          16.367331               14.80912
## Archer Heights        9.863461               14.80912
## Avondale             40.533851               14.80912
## Beverly              21.046464               14.80912
## Bridgeport           13.304517               14.80912
## Brighton Park        28.548742               14.80912
## Burnside             12.741349               14.80912
## Calumet Heights      11.465091               14.80912
```

Estimated random effects

```
coef(m1)
```

```
                (Intercept) overall_satisfaction
Albany Park        16.367331              14.80912
Archer Heights      9.863461              14.80912
Avondale           40.533851              14.80912
Beverly            21.046464              14.80912
Bridgeport         13.304517              14.80912
```

> What is the same for every neighborhood?

Estimated random effects

```
coef(m1)
```

```
                (Intercept) overall_satisfaction
Albany Park        16.367331              14.80912
Archer Heights      9.863461              14.80912
Avondale           40.533851              14.80912
Beverly            21.046464              14.80912
Bridgeport         13.304517              14.80912
```

> What is the same for every neighborhood?

The slope ($\widehat{\beta}_1$)

Estimated random effects

```
coef(m1)
```

```
                (Intercept) overall_satisfaction
Albany Park        16.367331              14.80912
Archer Heights      9.863461              14.80912
Avondale           40.533851              14.80912
Beverly            21.046464              14.80912
Bridgeport         13.304517              14.80912
```

> What is *different* for each neighborhood?

Estimated random effects

```
coef(m1)
```

```
                (Intercept) overall_satisfaction
Albany Park       16.367331              14.80912
Archer Heights     9.863461              14.80912
Avondale          40.533851              14.80912
Beverly           21.046464              14.80912
Bridgeport        13.304517              14.80912
```

> What is *different* for each neighborhood?

The intercept ($\widehat{\beta}_0 + \widehat{u}_i$)

Estimated random effects

	(Intercept)	overall_satisfaction
Albany Park	16.367331	14.80912
Archer Heights	9.863461	14.80912
Avondale	40.533851	14.80912
Beverly	21.046464	14.80912
Bridgeport	13.304517	14.80912

> How do I get the random effect estimates $\widehat{u}_i$?

Estimated random effects

	(Intercept)	overall_satisfaction
Albany Park	16.367331	14.80912
Archer Heights	9.863461	14.80912
Avondale	40.533851	14.80912
Beverly	21.046464	14.80912
Bridgeport	13.304517	14.80912

> How do I get the random effect estimates $\widehat{u}_i$?

The intercept for each neighborhood is $\widehat{\beta}_0 + \widehat{u}_i$

So subtract the estimate $\widehat{\beta}_0$ from the intercept for each neighborhood

Checking assumptions

$$Price_{ij} = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

- **Normality assumption:** Both $u_i \sim N(0, \sigma_u^2)$ and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

> How do you think we could check the normality assumption?

QQ plots

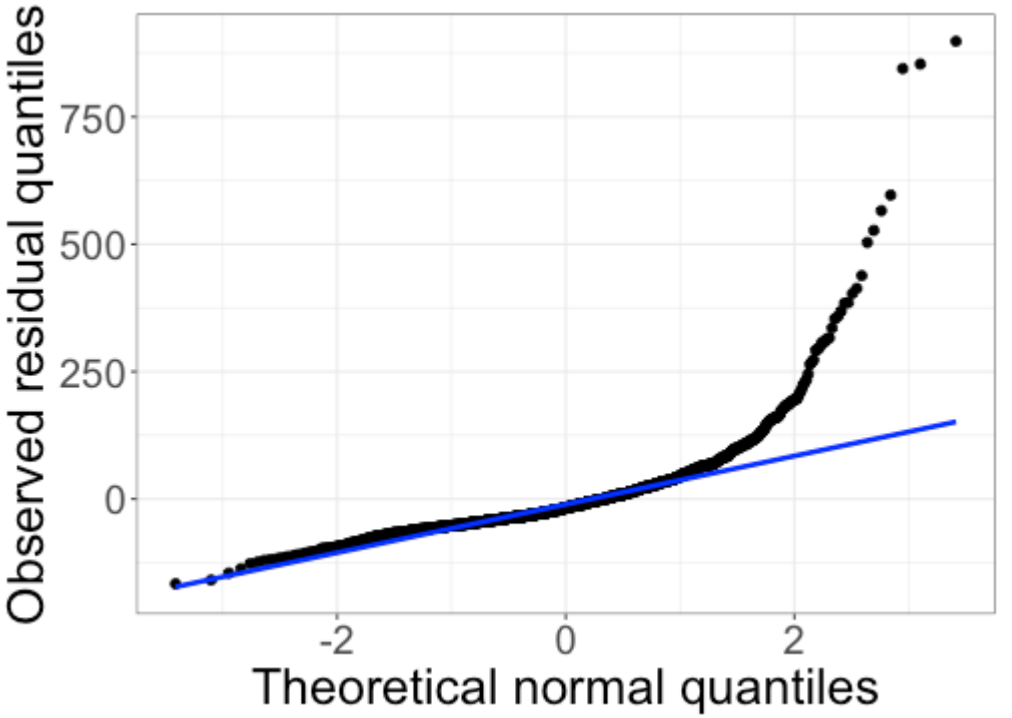
Assumption: $u_i \sim N(0, \sigma_u^2)$

- Check whether the random effect estimates $\widehat{u}_i$ appear normal with a QQ plot

Assumption: $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$

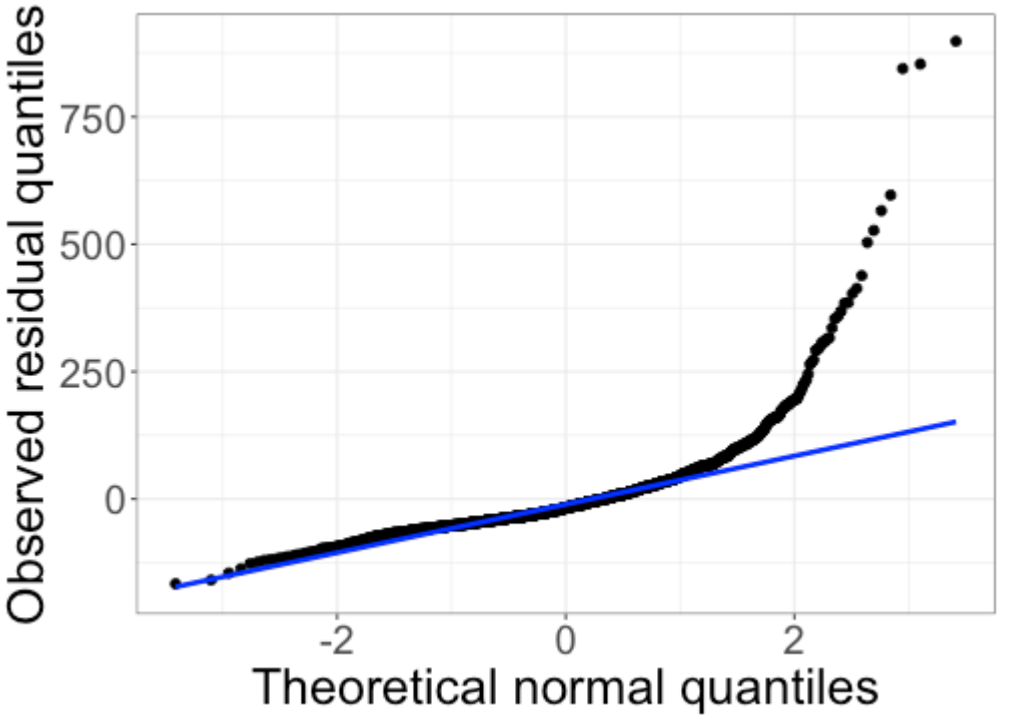
- Check whether the residuals appear normal with a QQ plot

QQ plot for the residuals



Do the residuals appear normal?

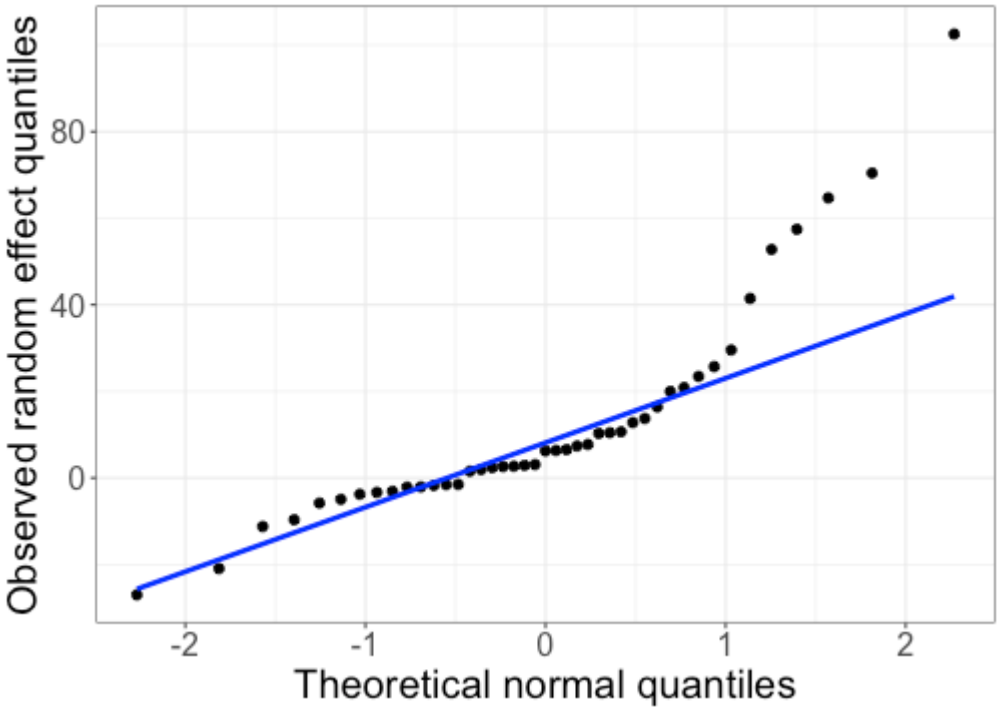
QQ plot for the residuals



> Do the residuals appear normal?

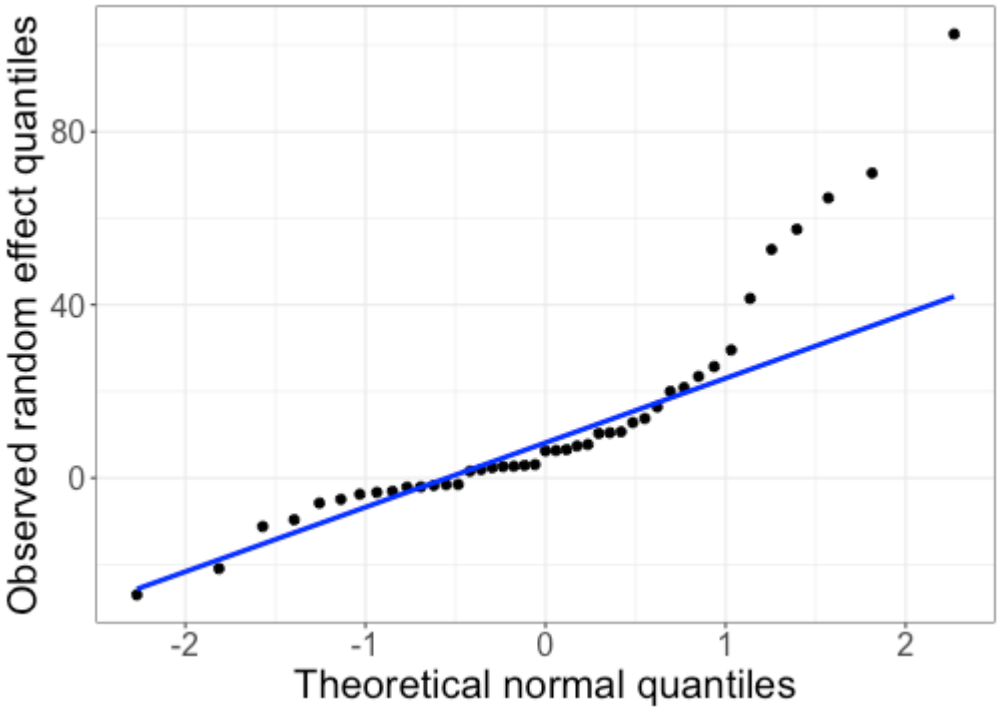
No -- though the sample size is large, so it might not matter

QQ plot for the random effects



Do the random effects appear normal?

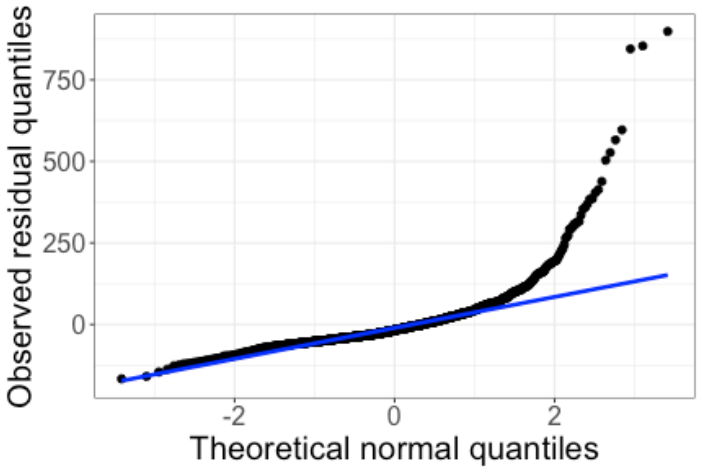
QQ plot for the random effects

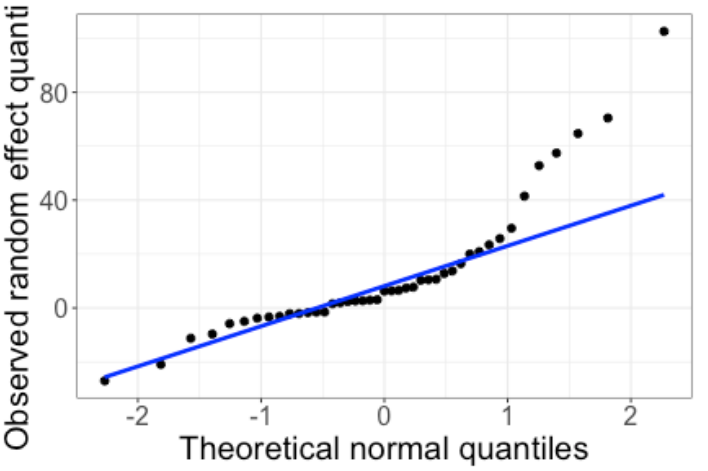


> Do the random effects appear normal?

No -- and this could be more problematic

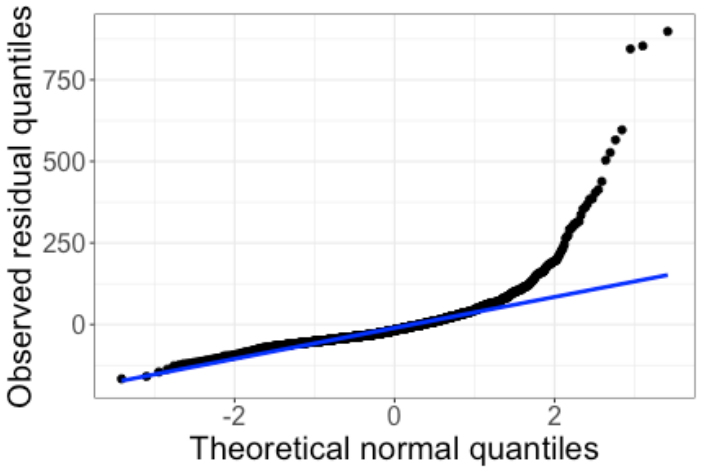
Addressing assumption violations

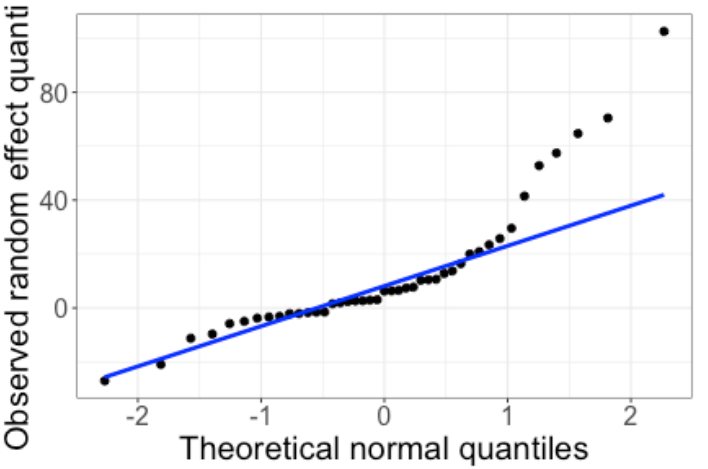




How could we address violations of the normality assumptions?

Addressing assumption violations





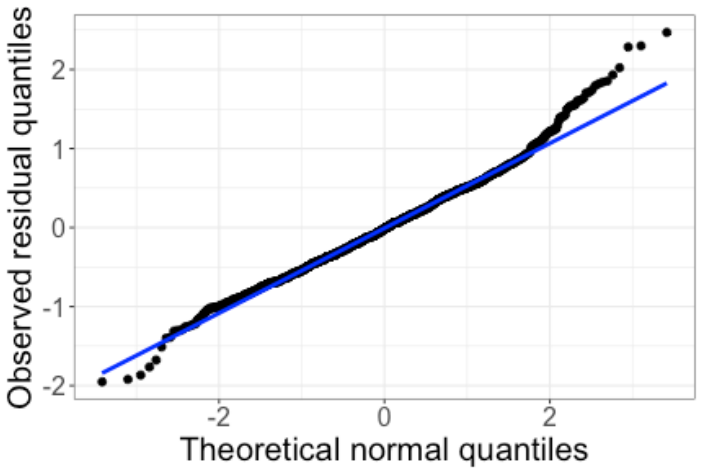
> How could we address violations of the normality assumptions?

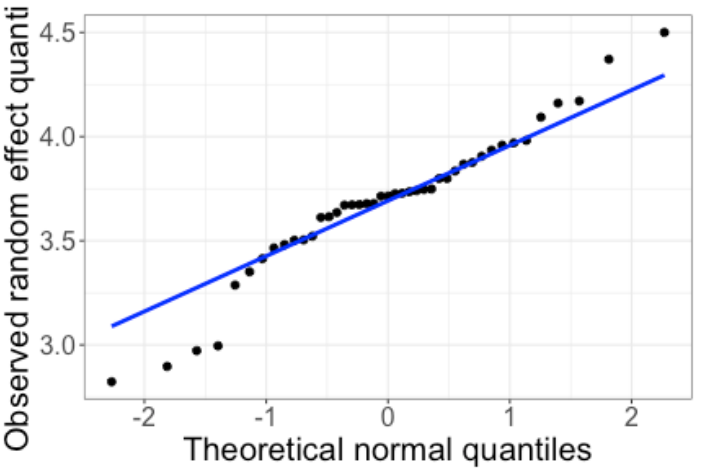
Transformations!

Transformations

$$\log(Price_{ij}) = \beta_0 + \beta_1 Satisfaction_{ij} + u_i + \varepsilon_{ij}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$





- Transformations can help with both the residuals and random effects

Class activity

Work on the class activity on the course website. Submit your HTML on Canvas when finished.